Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/26/2025
PI Name:	Weaver, Ashley Ph.D.		
Project Title:	Quantitative CT and MRI-based Modeling Assessment of Dynamic Vertebral Strength and Injury Risk Following Long-Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Winston-Salem	**State:**	NC
Zip Code:	27101-4101	**Congressional District:**	12
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	10/01/2016	**End Date:**	08/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/31/2025 per NSSC information (Ed., 7/12/24) NOTE: End date changed to 8/31/2024 per S. Mack-Phillips/JSC (Ed., 7/18/23) NOTE: End date changed to 8/31/2023 per V. Lehman/JSC (Ed., 3/29/23) NOTE: End date changed to 8/31/2022 per L. Barnes-Moten/JSC (Ed., 8/2/21) NOTE: End date changed to 8/31/2021 per NSSC information/S. Huppman/HRP (Ed., 2/25/2020) NOTE: End date changed to 2/1/2020 per NSSC information (Ed., 7/8/19)		
Key Personnel Changes/Previous PI:	None		

COI Name (Institution):	Stitzel, Joel Ph.D. (Wake Forest University School of Medicine) Tooze, Janet Ph.D. (Wake Forest University School of Medicine) Devane, Karan (Wake Forest University School of Medicine)
Grant/Contract No.:	NNX16AP89G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Prolonged periods of near weightlessness can cause damage to astronauts' musculoskeletal system. This damage can increase the risk of skeletal tissue failure (e.g., fractures, tears) when experiencing forceful, dynamic loads. Fractures of the spine during dynamic conditions such as launch or landing could cause a mission to fail. This study will measure this degradation of astronauts' vertebrae and spinal muscles during missions aboard the International Space Station (ISS). We will then determine the extent of vertebral weakening of crewmembers during long-duration missions. Changes in pre- and post-flight vertebral geometry, volume, and bone mineral density will be measured from existing lumbar quantitative computed tomography (qCT) scans, as well as from planned qCT scans of the cervical, thoracic, and lumbar spine from nine ISS crewmembers. Likewise, the pre- and post-flight spinal muscle volumes will be analyzed using both existing magnetic resonance imaging (MRI) scans and planned MRI scans from nine ISS crewmembers. The qCT and MRI scans will be analyzed to determine structural and material changes in the cervical, thoracic, and lumbar vertebrae and the spinal muscles that indicate damage which could weaken these tissues. Our unique engineering approach will measure the loss of vertebral strength during spaceflight conditions and predict the risk of failure during traumatic, dynamic loading conditions such as launch or landing. Vertebral strength and risk for vertebral fracture and injury will be quantified in dynamic simulations using a full human body model that is constructed using structural and material data gathered from the pre- and post-flight medical images for each astronaut. This study has significance in quantifying and addressing risks of long-duration spaceflight, including vertebral injury from dynamic loads, vertebral fracture, early onset vertebral osteoporosis due to spaceflight, and impaired performance due to reduced spinal muscle mass, strength, and endurance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Microgravity induces similar spinal changes to those seen in the aging population and people with limited mobility. Demonstrating how the vertebral column changes in response to microgravity can aid in refining the diagnostic and treatment protocols of physicians on Earth. Additionally, assessing vertebral column strength using finite element modeling can provide future techniques for assessing the efficacy of osteoporosis treatments, which would particularly benefit older adults.
Task Progress:	Below we summarize progress on the objectives of this project for the prior reporting year. This reporting period focused on: 1) integrating pre-flight and post-flight bone and muscle data collected from the astronaut medical imaging into subject-specific full human body models, and 2) conducting finite element simulations with those subject-specific models. Previously pre-flight and post-flight quantitative computed tomography (qCT) and magnetic resonance imaging (MRI) scans were collected from nine International Space Station crewmembers. The qCT scans were analyzed to measure vertebral bone mineral density, area, volume, and dimensions (height, length, and width). The MRI scans were analyzed to measure the neck and thoracolumbar muscle areas. The geometry and material properties were adjusted in a full human body model to match the pre- versus post-flight anatomy (including vertebral bone mineral density and spinal musculature) of each of the nine crewmembers. Along with vertebral dimensions (height, length, and width), the crewmember anthropometric measurements were used as inputs to morph the outer surface geometry, and the internal anatomy (including the spine) of a full human body model. The material properties of the vertebrae in the subject-specific full human body models were tuned by calibrating the elastic modulus of vertebral elements to the pre-flight or post-flight bone mineral density of the crewmember measured in the qCT scans. Physiological cross-section areas of the active musculature in the human body models were tuned for each subject (pre- and post-flight) using their muscle area data from the MRI scans. Thus, the resulting subject-specific models represented the crewmember's external anthropometry and vertebral size, vertebral bone mineral density, and muscle physiological cross-section areas values with separate models representing the: 1) pre-flight crewmember anatomy, and 2) the post-flight crewmember anatomy. Using these subject-specific models, simulations of landing scenarios (representative of loading conditions that can be experienced in spacecraft landing events upon return to Earth) were conducted to determine subject-specific assessments of forces, moments, and injury metrics in the cervical, thoracic, and lumbar spine. A test matrix of over 600 finite element simulations was conducted varying the timepoint (pre-flight vs. post-flight for each subject), loading direction (e.g., rear, frontal, vertical, rear/vertical, frontal/vertical, lateral/vertical), pulse magnitude (5-15 g), and pulse shape (direct vs. indirect). Injury metrics output from the simulations included neck forces (compression and tension), neck bending moments (flexion, extension, and lateral), the Neck Injury Criterion (Nij), and compression forces in the thoracolumbar vertebrae. These metrics were compared to published Injury Assessment Reference Values for injury tolerance in spaceflight scenarios.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Weaver AA, Mumtaz M, Thrush K, Jeong T, Devane K. "Impact of microgravity on astronauts' vertebral strength." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025

Abstracts for Journals and Proceedings	Park J, Devane K, Busquets TE, Madrid D, Poveda L, Taylor A, Stitzel JD, Weaver AA. "Personalizing an average male human body model with MRI-derived muscle data from long-duration astronauts." 2025 Annual Meeting of the Biomedical Engineering Society, San Diego, California, October 8-12, 2025. Abstracts. 2025 Annual Meeting of the Biomedical Engineering Society, San Diego, California, October 8-12, 2025. , Oct-2025
Abstracts for Journals and Proceedings	Park J, Devane K, Busquets TE, Madrid D, Poveda L, Taylor A, Stitzel JD, Weaver AA. "Personalizing an average male human body model with MRI-derived muscle data from long-duration astronauts." Gulf Coast Undergraduate Research Symposium, Houston, Texas, October 18-19, 2025. Abstracts. Gulf Coast Undergraduate Research Symposium, Houston, Texas, October 18-19, 2025. , Oct-2025
Articles in Peer-reviewed Journals	Poveda L, Dash S, Madrid D, Devane K, Lenchik L, Tooze J, Weaver AA. "Thoracolumbar spine muscle size and composition changes in long-duration space missions." Life Sci Space Res (Amst). 2025 Feb;44:1-8. https://doi.org/10.1016/j.lssr.2024.11.003 ; PMID: 39864901 , Feb-2025